

Student Dropout Analysis

Exploratory Data Analysis (EDA) on Student Academic and Demographic Patterns

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Domain: Education Analytics

Tools: Python • Pandas • Matplotlib

Project Overview

This project performs an comprehensive **Exploratory Data Analysis (EDA)** on student academic, socioeconomic, and demographic data to identify key indicators and patterns associated with student retention and academic success. The analysis specifically investigates three primary student outcomes: **Graduate**, **Dropout**, and **Enrolled**.

Implemented within a Python environment (Jupyter Notebook / Google Colab), the project demonstrates an end-to-end analytical workflow spanning data loading, data quality audits, feature inspection, target distribution analysis, and multivariate visualization.

Project Objectives

- **Dataset Structure & Inspection:** Inspect the structural integrity and dimensionality of the dataset.
- **Data Quality Assessment:** Audit feature data types and identify missing or irregular values.
- **Variable Exploration:** Examine unique values, academic attributes, and demographic factors.
- **Target Analysis:** Quantify the distribution of graduation, dropout, and enrollment statuses.
- **Multivariate Evaluation:** Contrast academic, socioeconomic, and institutional variables across student outcomes.
- **Pattern Visualization:** Synthesize descriptive findings into visual distributions to highlight dropout risks.
- **Applied Skills Showcase:** Demonstrate foundational Python programming and structured data analysis capabilities.

Dataset Summary

The dataset captures multidimensional information regarding student cohorts, academic progression, and macroeconomic indicators.

ATTRIBUTE	DETAILS
Total Records	4,424 observations
Total Features	37 columns across academic, demographic, and economic dimensions
Target Variable	Target (Categorical)
Target Classes	Graduate, Dropout, Enrolled
Missing Values	No missing values reported across the dataset (100% complete)

Target Variable Distribution

The target distribution establishes the baseline outcome proportions among the 4,424 students evaluated:

STUDENT STATUS	NUMBER OF STUDENTS	PERCENTAGE (%)	PROPORTION REPRESENTATION
Graduate	2,209	49.93%	
Dropout	1,421	32.12%	
Enrolled	794	17.95%	

Variables Covered in Analysis

- **Demographic:** Marital status, Gender, Nationality, Age at enrollment, International student status.
 - **Background:** Parents' qualifications and occupations, Previous qualification and qualification grade.
 - **Institutional:** Application mode, Application order, Course, Attendance type (daytime/evening).
- **Financial:** Scholarship status, Debtor status, Tuition fee payment status.
 - **Academic Performance:** 1st and 2nd semester curricular units (credited, enrolled, evaluations, approved, grades).
 - **Macroeconomic Context:** Unemployment rate, Inflation rate, GDP growth.
 - **Outcome:** Student status (Target).

Tools & Technologies

TECHNOLOGY	ROLE & APPLICATION IN PROJECT
Python	Primary programming language for data pipeline execution and analysis.
Jupyter Notebook / Google Colab	Interactive development environments used for reproducible data science workflows.
Pandas	Data ingestion, structure auditing, transformations, aggregation, and grouping.
NumPy	High-performance numerical routines, array computations, and metrics support.
Matplotlib	Visual rendering of categorical distributions, trends, and feature relationships.

Analysis Workflow

1 Data Loading

Importing raw student dataset into a Python DataFrame environment for downstream operations.

2 Data Inspection

Auditing dataset dimensions (shape), column names, datatypes, sample head/tail records, and unique value cardinality.

3 Data Quality Analysis

Performing integrity checks for null/NaN elements, duplicate records, and anomalous schema entries.

4 Target Variable Analysis

Calculating frequency distributions and relative percentages across Graduate, Dropout, and Enrolled outcomes.

5 Exploratory Data Analysis (EDA)

Interrogating cross-tabulations and relationships among academic performance, financial solvency, and demographics.

6 Data Visualization

Generating comparative charts to effectively communicate patterns that correlate with student retention and dropout risks.

Repository Structure & Execution

Repository Structure

```
student-dropout-analysis/ | ├──  
Week_4_Task.ipynb |── README.md |  
requirements.txt
```

Note: requirements.txt specifies local execution dependencies.

How to Run

Option 1: Google Colab

1. Download or clone the project repository.
2. Upload Week_4_Task.ipynb directly to Google Colab.
3. Execute cells sequentially from top to bottom.

Option 2: Local Jupyter Notebook

1. Clone repository and install required packages.
2. Launch Jupyter Notebook within the repository folder.
3. Open Week_4_Task.ipynb and execute all cells.

Key Takeaways & Core Competencies

This project demonstrates an end-to-end Exploratory Data Analysis workflow utilizing real-world educational data. It translates raw operational indicators into actionable analytical summaries, establishing a clear profile of the factors impacting student completion rates.

Core Technical Competencies Demonstrated:

- | | |
|---|--|
| • Exploratory & Descriptive Data Analysis | • Data Inspection & Quality Assurance |
| • Python Data Science Stack (Pandas, NumPy) | • Target Distribution & Categorical Profiling |
| • Data Visualization with Matplotlib | • Jupyter & Google Colab Workflow Architecture |

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★ Feel free to explore the Jupyter Notebook ([Week_4_Task.ipynb](#)) for comprehensive code implementations, analytical breakdowns, and detailed charts.